Final project Data Science

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```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyr)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v readr     2.2.0
## v lubridate 1.9.5     v stringr   1.6.0
## v purrr     1.2.2     v tibble    3.3.1
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
gene_expression <- read.csv("~/Downloads/GSE85307_geneExpression.csv")
gene_names<- read.csv("~/Downloads/GSE85307_geneNames.csv")
participant_information<-read.csv("~/Downloads/GSE85307_covar-1.csv")


# Check if sample IDs match
sum(colnames(gene_expression)[-1] %in% participant_information$geoID)
```

```
## [1] 157
```

```r
# Check if probe IDs match gene names
sum(gene_expression$X %in% gene_names$PROBEID)
```

```
## [1] 500
```

```r
#  only samples that have matching participant information
matched_ids <- intersect(
  colnames(gene_expression)[-1],
  participant_information$geoID
)

# Select those samples and keep the probe ID column
expression_matched <- gene_expression[
  , c("X", matched_ids)
]

# Keep participant records in the same order as the expression samples
participant_matched <- participant_information[
  match(matched_ids, participant_information$geoID),
]

# Confirm the sample IDs are now in the same order
all(colnames(expression_matched)[-1] == participant_matched$geoID)
```

```
## [1] TRUE
```

```r
# Add the gene names to the expression data using probe IDs
gene_expression_named <- merge(
  gene_names,
  expression_matched,
  by.x = "PROBEID",
  by.y = "X"
)

# Check the result
head(gene_expression_named[, 1:3])
```

```
##   PROBEID    GENE GSM2264363
## 1 7897322   PHF13   7.197650
## 2 7897860 TNFRSF8   8.094866
## 3 7898167    CTRC   6.504210
## 4 7898653  FAM43B   9.015528
## 5 7898809   EPHB2   7.022149
## 6 7900468    NFYC   9.908033
```

```r
il24_row <- gene_expression_named[
  gene_expression_named$GENE == "IL24",
]

il24_row
```

```
##    PROBEID GENE GSM2264363 GSM2264379 GSM2264394 GSM2264418 GSM2264449
## 18 7909271 IL24   5.304722   5.345838   5.176851   5.012965   5.302342
##    GSM2264452 GSM2264362 GSM2264419 GSM2264416 GSM2264391 GSM2264502 GSM2264371
## 18   5.371454   5.165942   5.523268   5.200205   5.399988   5.141818    5.24976
##    GSM2264392 GSM2264395 GSM2264488 GSM2264445 GSM2264388 GSM2264430 GSM2264479
## 18   5.634554   4.950234   5.003732   5.070527   4.930634   5.429305   5.396054
##    GSM2264489 GSM2264476 GSM2264478 GSM2264447 GSM2264431 GSM2264367 GSM2264413
## 18   5.166491   5.327274   5.377547   4.946742   5.353672   5.292758   5.215476
##    GSM2264390 GSM2264417 GSM2264497 GSM2264455 GSM2264474 GSM2264470 GSM2264349
## 18   5.343375   5.249425   5.055451   5.363431   5.200656   5.032802   5.574995
##    GSM2264435 GSM2264501 GSM2264429 GSM2264385 GSM2264420 GSM2264496 GSM2264352
## 18   5.321304   5.664292   5.083202   5.294506   5.112094   5.382895    5.14918
##    GSM2264370 GSM2264408 GSM2264481 GSM2264397 GSM2264423 GSM2264412 GSM2264376
## 18   5.397628   4.985631   5.518786   5.355376   5.150704   5.368113   5.209312
##    GSM2264369 GSM2264498 GSM2264458 GSM2264448 GSM2264494 GSM2264386 GSM2264482
## 18   5.177185   5.084427    5.13668   5.290502   5.338092   5.427948    5.27907
##    GSM2264425 GSM2264406 GSM2264382 GSM2264396 GSM2264348 GSM2264358 GSM2264409
## 18   5.258006   5.184778   5.315045   5.336553   5.362432   5.281184   5.209363
##    GSM2264421 GSM2264347 GSM2264462 GSM2264437 GSM2264463 GSM2264483 GSM2264499
## 18   5.329229    5.17608   5.251802   5.126546    5.12512   5.135645   5.186877
##    GSM2264353 GSM2264359 GSM2264490 GSM2264373 GSM2264457 GSM2264407 GSM2264500
## 18   5.116099   5.451734   5.368263   5.235994   5.133509   5.276181   5.277204
##    GSM2264383 GSM2264450 GSM2264377 GSM2264495 GSM2264451 GSM2264484 GSM2264467
## 18   5.331143   5.160008   5.199734   5.210348    5.29986   5.107351   5.444403
##    GSM2264472 GSM2264469 GSM2264410 GSM2264486 GSM2264460 GSM2264356 GSM2264491
## 18   5.337572   5.373479   5.189501   5.225794   5.119229   5.398464   5.349352
##    GSM2264453 GSM2264401 GSM2264439 GSM2264411 GSM2264357 GSM2264468 GSM2264487
## 18   5.213492   5.414249   5.737155    5.16917   5.208745   5.359585    5.26906
##    GSM2264443 GSM2264399 GSM2264400 GSM2264403 GSM2264372 GSM2264480 GSM2264380
## 18     5.3381   5.199911   5.118983   5.626285   5.296868   5.231797   5.306509
##    GSM2264364 GSM2264360 GSM2264404 GSM2264393 GSM2264461 GSM2264374 GSM2264361
## 18   5.389025   5.153778   5.336674    5.22337   5.298961   5.214402   5.322069
##    GSM2264405 GSM2264459 GSM2264424 GSM2264381 GSM2264446 GSM2264454 GSM2264389
## 18   5.210049   5.156236   5.393652   5.211188    5.34157   5.372299   5.453933
##    GSM2264432 GSM2264368 GSM2264492 GSM2264465 GSM2264384 GSM2264485 GSM2264434
## 18   5.216978   5.578438   5.361869   4.986148   5.176326   5.386441   5.345505
##    GSM2264471 GSM2264426 GSM2264415 GSM2264398 GSM2264456 GSM2264402 GSM2264442
## 18   5.174611   5.111049   5.338892   5.146762    5.26826   5.246893   5.268122
##    GSM2264440 GSM2264433 GSM2264378 GSM2264387 GSM2264375 GSM2264441 GSM2264444
## 18   5.462637   5.300272   5.066712   5.403466   5.219215   5.091278   5.276951
##    GSM2264477 GSM2264464 GSM2264350 GSM2264355 GSM2264466 GSM2264422 GSM2264366
## 18   5.476952   5.072869   5.141216    5.28812   5.197534    5.32169   5.126709
##    GSM2264354 GSM2264414 GSM2264436 GSM2264351 GSM2264493 GSM2264428 GSM2264365
## 18   5.295288   5.139097   5.396007   5.366565   5.306501   5.342107   5.192101
##    GSM2264346 GSM2264427 GSM2264473 GSM2264475 GSM2264438
## 18   5.402753   5.279991   5.065054   5.169129   5.244771
```

```r
# Select IL24 gene expression
il24_data <- gene_expression_named %>%
  filter(GENE == "IL24")

# Convert gene expression data to long format
il24_long <- il24_data %>%
  select(-PROBEID, -GENE) %>%
```

```
  pivot_longer(
    cols = everything(),
    names_to = "geoID",
    values_to = "IL24_expression"
  )

# Match IL24 expression to participant information
il24_analysis <- il24_long %>%
  left_join(participant_matched, by = "geoID")

# Check the data
head(il24_analysis)
```

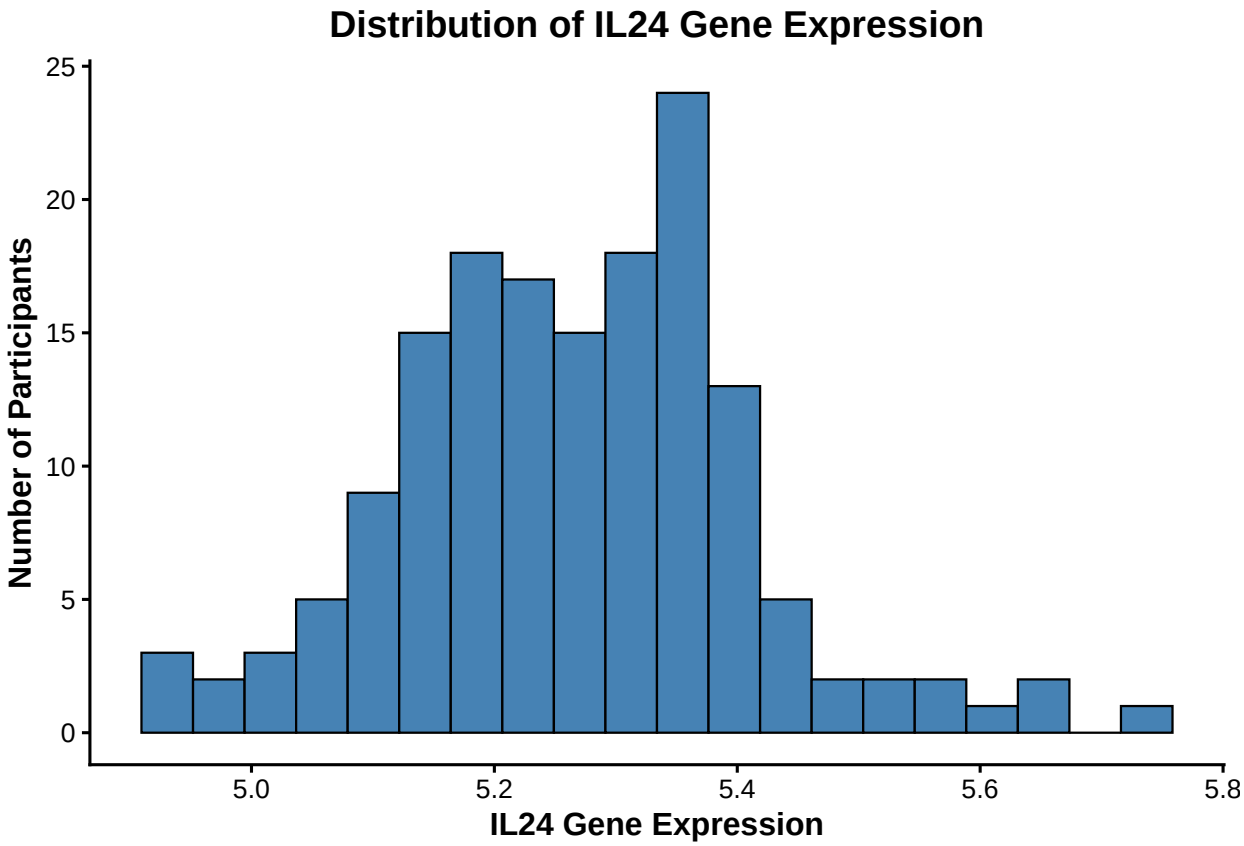
```
## # A tibble: 6 x 11
##   geoID      IL24_expression subjectID mat_age_yrs            preg_preeclampsia
##   <chr>                <dbl> <chr>     <chr>                  <chr>
## 1 GSM2264363            5.30 8054547-M age_years: 33.09514032 pregnancy_conditi~
## 2 GSM2264379            5.35 6051478-M age_years: 26.56536619 pregnancy_conditi~
## 3 GSM2264394            5.18 1090275-M age_years: 28.43531828 pregnancy_conditi~
## 4 GSM2264418            5.01 1075381-M age_years: 20.87611225 pregnancy_conditi~
## 5 GSM2264449            5.30 1084308-M age_years: 18.87474333 pregnancy_conditi~
## 6 GSM2264452            5.37 6017807-M age_years: 20.86516085 pregnancy_conditi~
## # i 6 more variables: delivery_gestAge_wks <chr>,
## #   sampleCollection_gestAge_wks <chr>, batch_Microarray <chr>, mat_BMI <chr>,
## #   mat_vitD <chr>, mat_asthma <chr>
```

```
nrow(il24_analysis)
```

```
## [1] 157
```

```
# Histogram of IL24 gene expression
plot1 <- ggplot(il24_analysis, aes(x = IL24_expression)) +
  geom_histogram(
    bins = 20,
    fill = "steelblue",
    color = "black",
    linewidth = 0.4
  ) +
  labs(
    title = "Distribution of IL24 Gene Expression",
    x = "IL24 Gene Expression",
    y = "Number of Participants"
  ) +
  theme_classic(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.title = element_text(face = "bold")
  )

# Display histogram
print(plot1)
```




```
# PLOT 2: SCATTERPLOT
# IL24 Gene Expression vs. Maternal Age
# Convert maternal age to numeric
il24_analysis$mat_age_yrs <- as.numeric(
  sub(".*: ", "", as.character(il24_analysis$mat_age_yrs))
)

# Check the result
str(il24_analysis$mat_age_yrs)
```

```
##  num [1:157] 33.1 26.6 28.4 20.9 18.9 ...
```

```
summary(il24_analysis$mat_age_yrs)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   18.43   22.15   26.57   26.38   30.14   38.76
```

```
plot2 <- ggplot(il24_analysis,
                aes(x = mat_age_yrs,
                    y = IL24_expression)) +
  # Add dusty rose points
  geom_point(size = 2, alpha = 0.7, color = "#C06C84") +

  # Add a lavender smooth trend line
```

```r
  geom_smooth(method = "loess", se = TRUE,
              color = "#765285", fill = "#D9C2E9") +

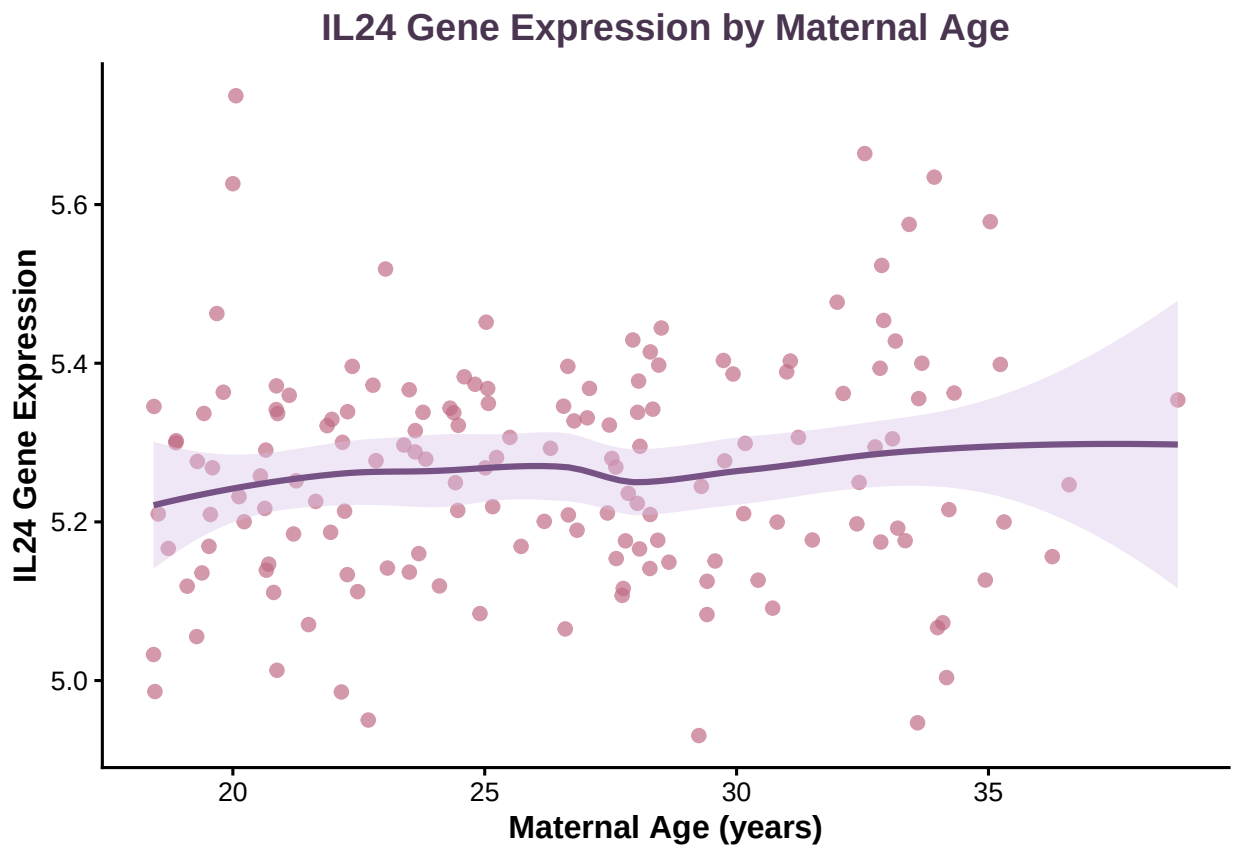
  # Add labels
  labs(
    title = "IL24 Gene Expression by Maternal Age",
    x = "Maternal Age (years)",
    y = "IL24 Gene Expression"
  ) +

  # Use a clean theme
  theme_classic(base_size = 12) +

  # Format the title and axis labels
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5,
                              color = "#49354F"),
    axis.title = element_text(face = "bold")
  )

# Display the scatterplot
print(plot2)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```r
# PLOT 3: BOXPLOT
# IL24 Expression by Preeclampsia Status and BMI

# Extract the numeric BMI values from the original text
il24_analysis$mat_BMI <- as.numeric(
  sub(".*: ", "", il24_analysis$mat_BMI)
)

# Check that the conversion worked
summary(il24_analysis$mat_BMI)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   17.95   22.90   27.33   28.94   33.27   61.95
```

```r
# Create BMI categories
il24_analysis <- il24_analysis %>%
  mutate(
    BMI_category = cut(
      mat_BMI,
      breaks = c(-Inf, 18.5, 25, 30, Inf),
      labels = c(
        "Underweight",
        "Normal weight",
        "Overweight",
        "Obesity"
      ),
      right = FALSE
    )
  )

# Check the number of participants in each category
table(il24_analysis$BMI_category, useNA = "ifany")
```

```
##
##   Underweight Normal weight    Overweight       Obesity
##             5            48            47            57
```

```r
# Convert maternal BMI to numeric
il24_analysis$mat_BMI <- as.numeric(
  as.character(il24_analysis$mat_BMI)
)

# Check the BMI variable
str(il24_analysis$mat_BMI)
```

```
##  num [1:157] 18.4 18.4 21.4 25.9 27.9 ...
```

```r
# Create BMI categories
il24_analysis <- il24_analysis %>%
  mutate(
    BMI_category = cut(
```

```
      mat_BMI,
      breaks = c(-Inf, 18.5, 25, 30, Inf),
      labels = c(
        "Underweight",
        "Normal weight",
        "Overweight",
        "Obesity"
      ),
      right = FALSE
    )
  )

plot3 <- ggplot(
  il24_analysis,
  aes(
    x = preg_preeclampsia,
    y = IL24_expression,
    fill = BMI_category
  )
) +
  geom_boxplot(
    position = position_dodge(width = 0.8),
    na.rm = TRUE,
    color = "#51445F",
    linewidth = 0.4,
    alpha = 0.9
  ) +
  scale_fill_manual(
    values = c(
      "Underweight" = "#B8A1D9",
      "Normal weight" = "#E8A0BF",
      "Overweight" = "#E9B872",
      "Obesity" = "#8AB9B0"
    )
  ) +
  labs(
    title = "IL24 Expression by Preeclampsia Status and BMI",
    x = "Pregnancy Outcome",
    y = "IL24 Gene Expression",
    fill = "Maternal BMI Category"
  ) +
  theme_classic(base_size = 12) +
  theme(
    plot.title = element_text(
      face = "bold",
      hjust = 0.5,
      color = "#49354F"
    ),
    axis.title = element_text(face = "bold"),
    axis.text = element_text(color = "#333333"),
    legend.position = "right",
    legend.title = element_text(face = "bold")
  )
```

```r
# Display the updated boxplot
print(plot3)
```

